Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 0.1 | **Date:** 06-Apr-2026

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1. Administrative & Study Identification

Full Official Study Title: A Phase 3, Randomized, Open-label, Active-Comparator-Controlled Clinical Study to Evaluate the Safety and Efficacy of Bomedemstat (MK-3543/IMG-7289) Versus Best Available Therapy (BAT) in Participants With Essential Thrombocythemia Who Have an Inadequate Response to or Are Intolerant of Hydroxyurea **Study Title:** A Study of Bomedemstat (IMG-7289/MK-3543) Compared to Best Available Therapy (BAT) in Participants With Essential Thrombocythemia and an Inadequate Response or Intolerance of Hydroxyurea (MK-3543-006) **Short Title/ Acronym:** SHORESPAN (MK-3543-006) **Protocol Number:** MK-3543-006 / IMG-7289-CTP-301 **NCT Number:** NCT06079879 **Posting ID:** 1015841870201971649 **Sponsor/Company Name:** Merck **Investigational Product:** Bomedemstat (MK-3543 / IMG-7289) **Phase of Development:** Phase 3 **Trial Status:** Recruiting **Medical Monitor:** Merck Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether bomedemstat is superior to the best available therapy (BAT) with respect to achieving a durable clinicohematologic response (DCHR). **Secondary Objectives:** To evaluate the safety, pharmacokinetics, change from baseline in fatigue and symptom scores (via MFSAF v4.0 and PROMIS Fatigue SF-7a), durable reduction of platelet and white blood cell (WBC) counts, and reduction of mutation burden.

2.2 Background & Rationale Essential Thrombocythemia (ET) is a myeloproliferative neoplasm (MPN) driven by mutations (JAK2, CALR, MPL) characterized by thrombocytosis and megakaryocyte hyperplasia. While current treatments can prevent thrombotic complications, they do not substantially alter the disease's natural history. Lysine-specific demethylase 1 (LSD1) regulates hematopoietic stem and progenitor cell proliferation and is overexpressed in MPNs. In Phase 2 studies, the LSD1 inhibitor bomedemstat improved symptoms, durably reduced cell counts, and lowered mutation burdens. This study is being conducted to offer a new therapeutic option for ET patients who have an inadequate response to or are intolerant of standard hydroxyurea.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active-Comparator-Controlled (Best Available Therapy: anagrelide, busulfan, interferon alfa/pegylated interferon, or ruxolitinib) **Blinding:** Open-label **Randomization:** 1:1, Stratified by hydroxyurea history (inadequate response vs. intolerance) and MFSAF v4.0 baseline score (≥ 4 vs. < 4)

3.2 Planned Enrollment & Duration Targeted Enrollment (N): 340 evaluable patients **Number of Sites:** Global multicenter trial **Estimated Study Start:** 31-Dec-2023 **Estimated Primary Completion:** 30-Jul-2027

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Has a diagnosis of ET per WHO 2016 diagnostic criteria for myeloproliferative neoplasms (confirmed by a central pathologist). 3. Has a centrally assessed bone marrow fibrosis score of Grade 0 or Grade 1, as per a modified version of the European Consensus Criteria for Grading Myelofibrosis. 4. Has a platelet count $> 450 \times 10^9/L$ (450k /μL) and an absolute neutrophil count (ANC) $\ge 0.75 \times 10^9/L$ assessed up to 72 hours before first dose of study intervention. 5. Has a history of inadequate response to or intolerance of hydroxyurea based on modified European LeukemiaNet (ELN) criteria for hydroxyurea resistance or intolerance. 6. Has an inadequate or loss of response to their most recent prior ET therapy, requiring a change of cytoreductive therapy. 7. Participants may have received up to 3 prior ET-directed cytoreductive agents including hydroxyurea.

4.2 Exclusion Criteria 1. Known immediate or delayed hypersensitivity reaction or idiosyncrasy to drugs chemically related to bomedemstat or lysine demethylase or monoamine oxidase inhibitor (LSDi or MAOi) or the chosen best available therapy (including anagrelide, interferon alfa/pegylated interferon, ruxolitinib, or busulfan) that contraindicates participation. 2. History of any illness/impairment of GI function that might interfere with drug absorption (e.g., chronic diarrhea or history of gastric bypass surgical procedure), confound the study results or pose an additional risk to the individual by participation in the study. 3. Evidence at the time of Screening of increased risk of bleeding. 4. History of a malignancy, unless potentially curative treatment has been completed with no evidence of malignancy for 2 years. Note: The time requirement does not apply to participants who underwent successful definitive resection of basal cell carcinoma of the skin, squamous cell carcinoma of the skin, or carcinoma in situ, excluding carcinoma in situ of the bladder. 5. Human immunodeficiency virus (HIV)-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 6. Concurrent cytoreductive therapies outside the assigned treatment arm.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Bomedemstat 50 mg/day starting dose, titrated to a target platelet count of ≥ 150 to $\le 350 \times 10^9/L$. **Route of Administration:** Oral **Duration of Treatment:** Initial treatment phase of 52 weeks, with an extended treatment phase up to Week 156.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care. Patients in the BAT arm who have not discontinued trial treatment by Week 52 are eligible to crossover and receive bomedemstat during the extended phase. **Prohibited:** Concurrent investigational therapies or alternative cytoreductive therapies outside of the assigned study arm.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Bone Marrow Aspirate/Biopsy, Cheek Swab for Genomic Profiling
Baseline	Day 0	Randomization, First Dose, Baseline Symptom Questionnaires
Interim	Every 2 Weeks	Clinic visits every 2 weeks initially for dose titration, Safety Labs, Efficacy Assessment (Platelet/WBC counts)
End of Study	Up to Week 156	Final Vital Signs, Bone Marrow Biopsy/Aspirate review, Exit Interview, 30-day Safety Follow-up

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Durable Clinicohematologic Response (DCHR) Rate. **Measurement Method:** DCHR rate is the percentage of participants with DCHR, defined as a confirmed reduction of platelet count to $\le 400 \times 10^9/L$, absence of white blood cell (WBC) count elevation to $>10 \times 10^9/L$ locally assessed to be due to ET, and the absence of any thrombotic or major hemorrhagic events or disease progression to myelofibrosis (MF), myelodysplastic syndrome (MDS) or acute myeloid leukemia (AML) by Week 52.

7.2 Secondary & Exploratory Endpoints - Change From Baseline in Myelofibrosis Symptom Assessment Form (MFSAF) v4.0 Individual Fatigue Symptom Item Score. - Change From Baseline in Patient-reported Outcomes Measurement Information System (PROMIS) Fatigue SF-7a Total Fatigue Score. - Change From Baseline in Total Symptom Score as Measured on the MFSAF v4.0. - Reduction in molecular biomarkers / mutation burden. - Pharmacokinetics of bomedemstat.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via clinical visits (initially every 2 weeks) and patient symptom diaries. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness to the sponsor. **Data Monitoring Committee (DMC):** Independent External Data Monitoring Committee assessing ongoing safety and efficacy.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analysis; Safety Set for all participants receiving at least one dose of study medication. **Sample Size Justification:** $N = 340$ provides adequate power ($\alpha = 0.05$) to detect a statistically significant superiority of bomedemstat over Best Available Therapy (BAT). **Handling of Missing Data:** Multiple Imputation (MI) and classification of treatment discontinuations as non-responders for the primary DCHR endpoint.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote monitoring visits by MSD clinical research associates. **Audit Trail:** Validated eCRF systems with secure locking mechanisms, source data verification, and centralized data clarification forms.

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11. Study Identifiers & Governance

Primary ID: MK-3543-006 **Posting ID:** 1015841870201971649 **Registry Number**: NCT06079879 **Sponsor/Lead Organization**: Merck **Study Title**: A Study of Bomedemstat (IMG-7289/MK-3543) Compared to Best Available Therapy (BAT) in Participants With Essential Thrombocythemia and an Inadequate Response or Intolerance of Hydroxyurea (MK-3543-006) **Official Regulatory Title:** A Phase 3, Randomized, Open-label, Active-Comparator-Controlled Clinical Study to Evaluate the Safety and Efficacy of Bomedemstat (MK-3543/IMG-7289) Versus Best Available Therapy (BAT) in Participants With Essential Thrombocythemia Who Have an Inadequate Response to or Are Intolerant of Hydroxyurea

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12. Clinical Framework (Study Specific)

Disease Area / Primary Condition: Essential Thrombocythemia (ET) **Diagnosis Threshold:** Platelet count $> 450 \times 10^9/L$ **Medical Methodology:** Best Available Therapy (BAT) control vs. LSD1

Inhibition **Optimization Target:** Target platelet count of ≥ 150 to $\le 350 \times 10^9/L$

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13. Study Procedures & Methodology

Management Pathway: Targeted Cytoreductive Therapy Pathway for ET **Education Protocol (HCP):** Investigator protocol training on Bomedemstat dose titration algorithms **Education Protocol (Patient):** Self-reporting training for MFSAF v4.0 and PROMIS Fatigue SF-7a **Quality Audit Mechanism:** Centralized pathological review of bone marrow aspirates and biopsies

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14. Observation & Follow-Up Schedule

Total Duration: Up to 156 weeks **Onsite Visit Cadence:** Every 2 weeks initially, transitioning to less frequent maintenance intervals **Remote Monitoring Frequency:** 30 days post-last dose for final safety follow-up **Checklist Review Interval:** Routine symptom questionnaire (MFSAF v4.0) collection

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]